

Aziz Laadhar

Machine Learning & Quantitative Research | Scientific ML | Optimization

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RESEARCH PROFILE

EPFL M.Sc. Data Science candidate developing structure-aware machine-learning methods for high-dimensional temporal data, scientific inverse problems, and constrained decision systems, with research experience at Harvard and Schlumberger-Doll Research.

RESEARCH EXPERIENCE

Harvard University, Laboratory of Bioelectronics

Cambridge, MA
Jul 2026 – Present

Master's Thesis Researcher (Advisor: Prof. Jia Liu)

- Developing self-supervised representation-learning methods for animal-behavior video, with emphasis on masked-video pretraining and rigorous evaluation of transferable behavioral representations across heterogeneous datasets.
- Built a quantitative evaluation framework spanning eight behavior-analysis capability families, with held-out protocols, regression-tested failure modes, and reproducible diagnostics for comparing multimodal models under distribution shift.

SLB, Schlumberger-Doll Research

Cambridge, MA
Jul 2025 – Feb 2026

Machine Learning Researcher, Deep Sensing & Monitoring

- Developed a physics-guided encoder-decoder for 2D electromagnetic resistivity inversion, coupling predictions to a differentiable surrogate of a proprietary 2.5D forward simulator through model- and data-misfit losses; reduced mean-squared error 14.4%.
- Generated approximately 100,000 synthetic formations with faults and pinch-outs; benchmarked 6+ variants using edge-aware losses, noise augmentation, EMA, and systematic train-test distribution-shift analysis.

Sirius Energy SA

Lugano, Switzerland
Jul 2024 – Feb 2025

Quantitative Analyst, European Power & Gas Markets

- Modeled full hourly day-ahead supply and demand curves with functional autoregression and exogenous fundamentals, estimated as a second-order cone program in MOSEK with monotonicity constraints enforcing economically valid curve shapes.
- Integrated predicted curves into a European market-clearing simulator and built rolling backtests on implied clearing price and volume, connecting high-dimensional forecasts to production price, spread, and cross-border-flow scenarios.

EPFL, Visual Intelligence for Transportation Laboratory

Lausanne, Switzerland
2023 – 2024

Researcher, Learning for Constrained Combinatorial Optimization

- Implemented an 8-layer, 16-head graph Transformer and trained it with 50,000 Gurobi-generated expert instances plus PPO fine-tuning; reached within 2.67% of solver cost on held-out Dial-a-Ride instances while modeling capacity, time-window, and ride-time constraints.

SELECTED RESEARCH ARTIFACTS

The Transformer Network for the Dial-a-Ride Problem (equal contribution)

EPFL, 2024

Graph Transformer policy with imitation learning, PPO, greedy/beam decoding, and optimization baselines. [Code](#)

Deep Learning for Road Segmentation

EPFL, 2023

U-Net, LinkNet, and GC-DCNN experiments with augmentation and ablations; best test F1: 88.7%. [Code](#)

EPFL Mistral Mind Chatbot

EPFL, 2024

Domain-specific tutoring with retrieval-augmented generation, supervised fine-tuning, and direct preference optimization.

[Related code](#)

EDUCATION

École polytechnique fédérale de Lausanne (EPFL)

Lausanne, Switzerland
Sep 2023 – Dec 2026 (expected)

M.Sc. Data Science, Minor in Financial Engineering

Coursework: Advanced ML, Statistics, Optimization, Stochastic Calculus, Derivatives, Econometrics.

B.Sc. Communication Systems

Sep 2020 – Jul 2023

TEACHING & DISTINCTIONS

Winner, Mercuria-EPFL electricity-price forecasting competition | Teaching Assistant, Computer Architecture and Computer Networks (two years each)

TECHNICAL SKILLS

ML / research: PyTorch, deep learning, multimodal models, representation learning, reinforcement learning, graph neural networks, computer vision, inverse problems, experimental design

Quantitative: Time series, statistical inference, convex/conic optimization, numerical methods, market simulation, uncertainty-aware evaluation

Tools: Python, NumPy, pandas, scikit-learn, SQL, MATLAB, C++, Gurobi, MOSEK, Linux, Git, SLURM, GPU training